



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2007

MATHEMATICS

8:45 am – 11:50 am

Tuesday 28th August 2007

Directions to Students

• Reading Time : 5 minutes	• Total Marks 120
• Working Time : 3 hours	
• Write using blue or black pen. (sketches in pencil).	• Attempt Question 1 – 10
• Board approved calculators may be used	• All questions are of equal value
• A table of standard integrals is provided at the back of this paper.	
• All necessary working should be shown in every question.	
• Answer each question in the booklets provided and clearly label your name and teacher's name.	

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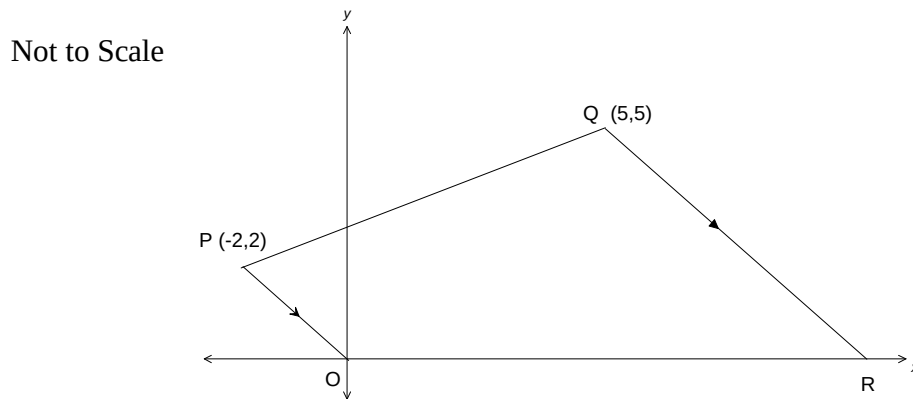
Total marks (120)
Attempt Questions 1 – 10
All questions are of equal value

Answer each question in a SEPARATE writing booklet.

QUESTION 1	(12 MARKS)	Use a SEPARATE writing booklet	Marks
(a)	Evaluate $\pi^{-1.6}$ correct to 2 significant figures.		2
(b)	Write down the exact value of		
	(i) $\operatorname{cosec} \frac{\pi}{6}$		1
	(ii) $10^{\log_{10} \sqrt{2}}$		1
(c)	Paul paid Jonestown mechanical repairs \$510.00 for his car to be serviced. This price included a 15% discount from the standard cost of the service. What was the standard cost of the service.		2
(d)	Factorise fully $2 - 54x^3$.		2
(e)	Solve $ y - 2 = 6$, for y .		2
(f)	Solve $5 - \frac{a}{9} < 3$, for a .		2

QUESTION 2**(12 MARKS)**

Use a SEPARATE writing booklet

Marks**(a)**

In the diagram, OPQR is a trapezium with OP parallel to RQ . The co-ordinates of O , P and Q are $(0, 0)$, $(-2, 2)$ and $(5, 5)$ respectively.

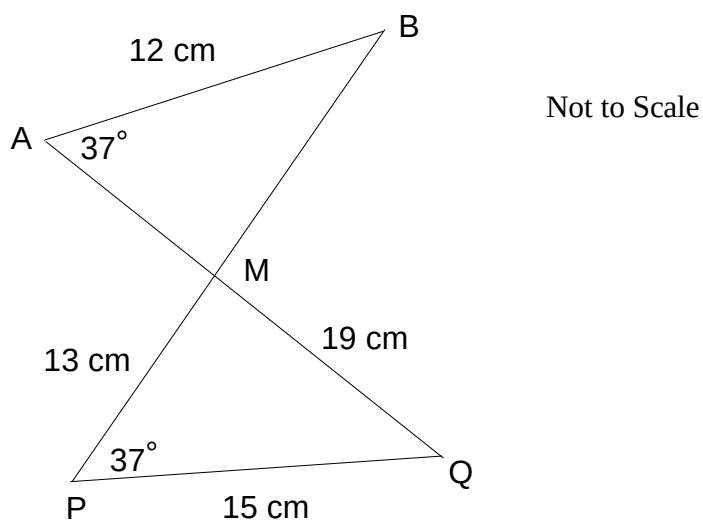
- | | | |
|-------|---|----------|
| (i) | Calculate the exact length of OP . | 1 |
| (ii) | Write down the gradient of OP . | 1 |
| (iii) | What is the size of angle POR . | 1 |
| (iv) | Find the equation of the line QR , and hence find the co-ordinates of R . | 2 |
| (v) | Show that the perpendicular distance from O to RQ is $5\sqrt{2}$ units. | 2 |
| (vi) | Hence, or otherwise, calculate the area of the trapezium $OPQR$. | 2 |
| (b) | Shade the region in the Cartesian plane for which the inequalities $y < x - 1$, $y \geq 0$ and $x \geq 4$ hold simultaneously. | 3 |

QUESTION 3 (12 MARKS) Use a SEPARATE writing booklet

Marks

(a)

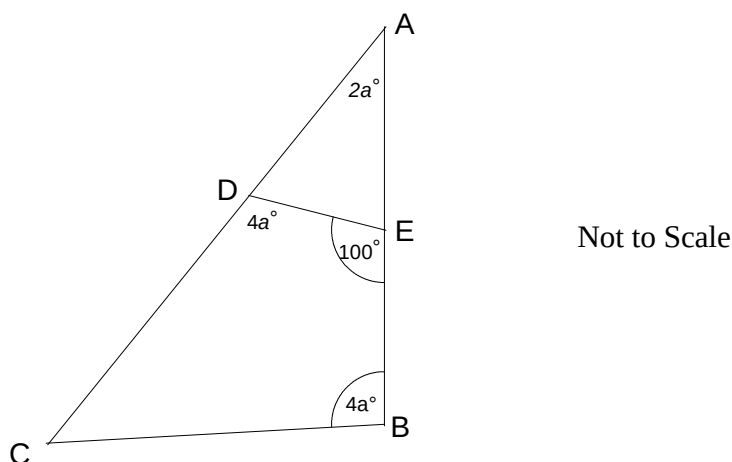
2



In the diagram, triangles BAM and QPM are similar. $BA = 12$ cm, $QP = 15$ cm, $QM = 19$ cm and $PM = 13$ cm. Find the length of side AM .

(b)

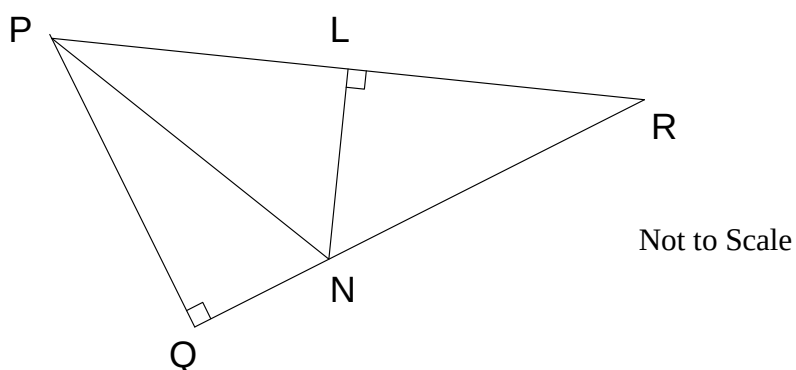
3



In the diagram ABC is a triangle and E and D are points on AB and AC respectively. Angle $CAB = 2a^\circ$, angle $EBC = \text{angle } CDE = 4a^\circ$ and angle $DEB = 100^\circ$. Find the value of a showing full working and give reasons.

QUESTION 3 (Continued)**Marks**

(c)



In the diagram PQR is a triangle with a right angle at Q. L is the midpoint of PR, and N is the point where the perpendicular to PR at L meets QR.

- (i) Show that the triangles PNL and RNL are congruent, giving reasons. **2**
- (ii) Suppose that it is also given that NP bisects the angle QPR. Find
- (α) the size of angle NRL **2**
- (β) the exact ratio LN : PR **2**
- (d) Write down the sum of the interior angles of a regular Heptagon (7 sided polygon). **1**

QUESTION 4 (12 MARKS) Use a SEPARATE writing booklet

Marks

(a) Differentiate with respect to x :

(i) $y = \log_e (3 + 2x^2)$. **1**

(ii) $y = (3e^x - 5)^7$. **2**

(iii) $y = \frac{\sin 3x}{x}$ **2**

(iv) $y = x\sqrt{x}$ **2**

(b) Find the equation of the tangent to the curve $y = -\frac{2}{x}$ at the point $(-1, 2)$. **3**

(c) Find $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 3x + 1}{2x^2 + 5} \right)$ **2**

QUESTION 5 (12 MARKS) Use a SEPARATE writing booklet **Marks**

- (a) The roots of the equation $2x^2 + 4x - 1 = 0$ are α and β .
- (i) Write down the values of $(\alpha + \beta)$ and $\alpha\beta$. **1**
- (ii) Evaluate $(\alpha^2 + \beta^2)$ **2**
- (b) Consider the quadratic equation $x^2 - (k - 2)x + (k + 1) = 0$, where 'k' is a constant.
- (i) Show that the discriminant is $(k^2 - 8k)$. **2**
- (ii) Find the values of 'k' for which the equation has real roots. **1**
- (c) A parabola has an equation in the form $x^2 - 12x = 8y - 52$
- (i) Express this equation in the form $(x - h)^2 = 4a(y - k)$ **1**
- (ii) Hence find the:
- (α) co-ordinates of the vertex **1**
- (β) co-ordinates of the focus **1**
- (γ) equation of the directrix **1**
- (d) Solve the equation $(5^x)^2 + 5^x - 2 = 0$, for x . **2**

QUESTION 6

(12 MARKS)

Use a SEPARATE writing booklet

Marks

(a) Find

$$(i) \int \frac{1}{4x^4} dx \quad 1$$

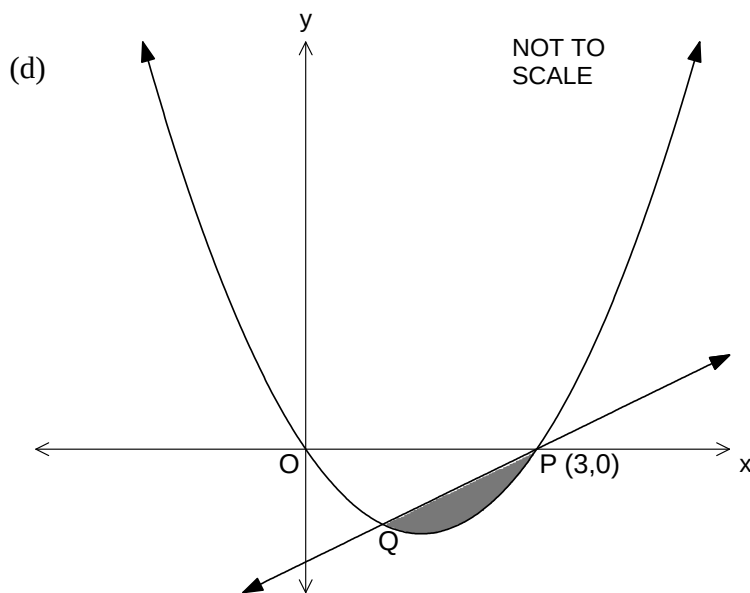
$$(ii) \int \frac{2x}{x^2+10} dx \quad 1$$

$$(b) \text{ Evaluate } \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2 x \, dx \quad 2$$

(c) Use Simpson's Rule with three function values to find an approximation for **3**

$$\int_3^7 \frac{x}{\ln x} dx.$$

Give your answer correct to one decimal place.



The graphs of $y = x - 3$ and $y = x^2 - 3x$ intersect at the points P (3,0) and Q, as shown in the diagram.

(i) Find the co-ordinates of Q. **2**(ii) Find the area of the shaded region bounded by $y = x^2 - 3x$ and $y = x - 3$. **3**

QUESTION 7

(12 MARKS) Use a SEPARATE writing booklet

Marks

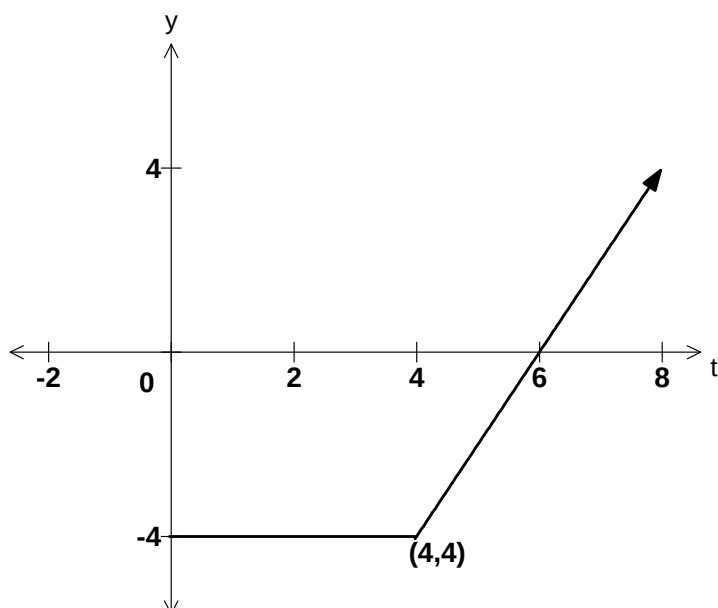
(a) Consider the curve $y = x^3 - 3x^2 + 3x - 1$.

(i) Show that the curve has only one stationary point, find its co-ordinates and determine its nature. **3**

(ii) State the values of x for which the curve is concave up. **1**

(iii) State the values of x for which the curve is increasing. **1**

(b)

**2**

The graph shows the function $y = f(x)$ whose domain is $x \geq 0$.

Trace or copy this graph onto your answer booklet

On the same axes, sketch the graph of the function $y = f'(x)$.

(c) The curve $y = f(x)$, $0 \leq x \leq 2\pi$ has a stationary point at $x = \frac{\pi}{4}$ and $f''(x) = \cos x + \sin x$. **3**

Find the x co-ordinate at which there is another stationary point.

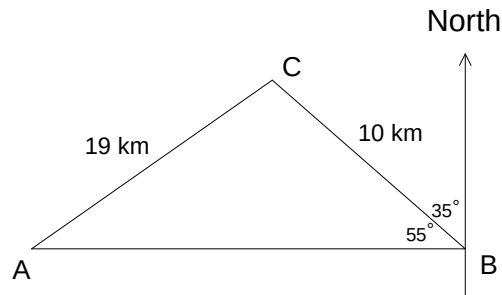
(d) If $y = x\sqrt{x+1}$, show that $\frac{dy}{dx} = \frac{3x+2}{2\sqrt{x+1}}$. **2**

QUESTION 8

(12 MARKS) Use a SEPARATE writing booklet

Marks

(a)



In the diagram, the point B is due east of point A. The point C is 19 km from point A and 10 km from point B. Point C is North 35° West of point B. Find the true bearing of point C from point A.

3

(b) The velocity of a particle, moving in a straight line, is given by

$$\frac{dx}{dt} = 2 - 4 \sin t \text{ for } 0 \leq t \leq 2\pi$$

where $\frac{dx}{dt}$ is measured in metres per second, t is measured in seconds and x in metres.

- (i) At what times during this period is the particle at rest. **2**
- (ii) Sketch the graph of velocity $\frac{dx}{dt}$ as a function of time for $0 \leq t \leq 2\pi$. **2**
(This sketch should be approximately half of one page)
- (iii) What is the maximum velocity of the particle during this period? **2**
- (iv) Calculate the exact total distance travelled by the particle **3**
between $t=0$ and $t=\frac{\pi}{2}$

QUESTION 9

(12 MARKS) Use a SEPARATE writing booklet

Marks

- (a) The population P of ants in a colony is determined by $P = 500 \times e^{\frac{t}{4}}$ where t is the time in weeks since the colony was originally established.
- (i) Find the size of the colony after 10 weeks. **1**
- (ii) How long would it take for the population to reach 10 000? **1**
- (iii) Find $\frac{dP}{dt}$ and explain what this means for the population trend? **2**
- (iv) Sketch the graph of $\frac{dP}{dt}$ against t . **2**
- (b) Paul and Wendy borrow \$20000 from the Miami Bank. This loan plus interest is to be repaid in equal monthly instalments of \$399 over five years. Interest of 7.2% p.a is compounded monthly on the balance owing at the start of each month.
- Let A_n be the amount owing after n months.
- (i) Over the five year repayment period, how much interest is charged? **1**
- (ii) Show that $A_1 = 19721$ **1**
- (iii) Clearly show that $A_2 = 20000 \times 1.006^2 - 399(1 + 1.006)$. **1**
- (iv) Deduce then that $A_n = 66500 - 46500 \times 1.006^n$ **2**
- (v) After two years of repayments Paul and Wendy decide on the very next day to repay the loan in one full payment. **1**
How much will this one payment be?

QUESTION 10 (12 MARKS) Use a SEPARATE writing booklet

Marks

- (a) The area bounded by the curve $y = \frac{1}{x}$, the x-axis and the ordinates $x = a$ and $x = 4$ is rotated about the x-axis. If the volume generated is $\frac{\pi}{2} \text{ unit}^3$, where $0 < a < 4$, find the value of a . **3**
- (b) If $x \sec \theta = y \tan \theta$, prove that $\tan \theta \sec \theta = \frac{xy}{y^2 - x^2}$ **3**
- (c) If the straight line $y = mx$ is a tangent to the curve $y = e^{\frac{x}{2}}$, find the exact value of m . Clearly show your working. **3**
- (d) The sum of the first 'n' terms of a series is given by $S_n = n^2 + 6n$. **3**
Prove that this series is arithmetic.

(Note: It will be insufficient to outline a proof which involves evaluating $S_1, S_2, \text{etc.}, T_1, T_2, \text{etc.}$)

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Solutions

B.L.C



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MATHEMATICS

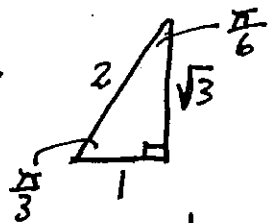
- This is a study resource for HSC preparation.
- This document is designed to help students understand the questions.
- The solutions should be treated as aids only.
- There may be better solutions to some questions.

Question 1

(a) $\pi^{-1.6} \approx 0.16$

From calculator
0.160162265...

(b) (i) $\operatorname{cosec} \frac{\pi}{6} = \frac{2}{1} = 2$



(ii) $10^{\log_{10} \sqrt{2}} = \sqrt{2}$ Rule: $a^{\log_a x} = x$

(a) IAW - Correct calcⁿ

IAW - Correct rounding

(i) IAW - Correct answer

(ii) IAW - Correct answer

(c) 85% of cost of standard service = \$510.00

1% of cost of standard service = $\$ \frac{510.00}{85}$

Cost of standard service = $\$ \left(\frac{510.00 \times 100}{85} \right)$

$= \$600.00$

IAW - Some attempt at unitary method

IAW - Correct answer

(d) $2 - 54x^3 = 2(1 - 27x^3)$

$= 2(1 - 3x)(1 + 3x + 9x^2)$

IAW - take out a factor of 2

IAW - correct answer

(e) $|y - 2| = 6$

$\pm(y - 2) = 6$

$y - 2 = 6$ or $y - 2 = -6$

$y = 8$ or $y = -4$

(e) Well done by most

IAW - $y = 8$ IAW - $y = -4$

(f) $5 - \frac{a}{9} < 3$

$-\frac{a}{9} < -2$

$a > 18$

(f) Many careless errors made here

IAW - Reasonable attempt

IAW - Correct answer

Question 2

(i) $O(0,0); P(-2,2)$

$$OP^2 = 2^2 + 2^2$$

$$OP^2 = 4 + 4$$

$$OP^2 = 8$$

$$OP = \sqrt{8} = 2\sqrt{2}$$

(i) Well answered

(ii) $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$m = \frac{2 - 0}{-2 - 0}$$

$$m = -1$$

(ii) well answered

(iii) $\tan \theta = m$

$$\tan \widehat{POR} = -1$$

$$\widehat{POR} = 135^\circ$$

(iii) many students did not know this result !!

(iv) $QR \parallel PO$

$$m_{QR} = m_{PO} = -1$$

Equation of QR:

$$y - y_1 = m(x - x_1)$$

using $A(5,5)$

$$y - 5 = -1(x - 5)$$

$$y - 5 = -x + 5$$

$$x + y - 10 = 0$$

Now let $y = 0$ for co-ordinates of R

$$\therefore x = 10$$

point $R = (10,0)$

(iv) many students did not read the second part of this question ----
→ Find the co-ord. of R

IAW - Equation of Line

IAW - Co-ord. of R

$$(v) \quad d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$BC: x + y - 10 = 0$$

Point $O(0,0)$

$$d = \frac{|1 \times 0 + 1 \times 0 - 10|}{\sqrt{1^2 + 1^2}}$$

$$d = \frac{10}{\sqrt{2}}$$

$$d = \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$d = 5\sqrt{2} \text{ units}$$

(v) Generally well answered

IAW - Substitution into Formula

IAW - Rationalising the denominator

(vi) Distance QR.

$$\overline{QR}^2 = (5-10)^2 + (5-0)$$

$$\overline{QR}^2 = (-5)^2 + 5^2$$

$$\overline{QR}^2 = 25 + 25$$

$$\overline{QR}^2 = 50$$

$$\overline{QR} = \sqrt{50} = 5\sqrt{2}$$

(vi) Many students had difficulty with this

IAW - Distance QR

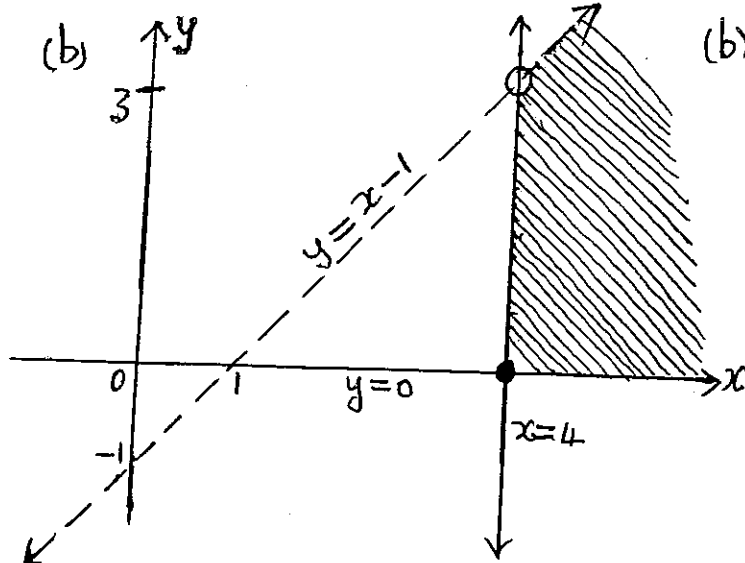
$$\text{Trapezium: } A = \frac{h}{2} (a+b)$$

$$A = \frac{5\sqrt{2}}{2} (2\sqrt{2} + 5\sqrt{2})$$

$$A = \frac{5\sqrt{2}}{2} \times 7\sqrt{2}$$

$$A = 35 \text{ units}^2$$

IAW - Use of Trapezium Formula



(b) Poorly answered by many

Marks were deducted for

1. Not showing intercepts
2. Not labelling diagram
3. Not showing dotted line for $y = x - 1$

Note - use a ruler
- use a pencil. 3

Question 3

(a) $\frac{AM}{13} = \frac{12}{15}$

$AM = \frac{12 \times 13}{15}$

$AM = 10.4 \text{ cm}$

(Note: Ratios of corresponding sides are equal)

(b) $\widehat{AED} = 180^\circ - 100^\circ = 80^\circ$

(Angle sum at E on a straight line AB is 180°)

$2a^\circ + 80^\circ = 6a^\circ$. External angle of $\triangle ADE$ equals the sum of the interior opposite angles.

$\therefore 4a = 80$

$a = 20$

Note There is more than one approach here.

(c)

(i) In the Δ 's $\triangle PNL$ and $\triangle RNL$

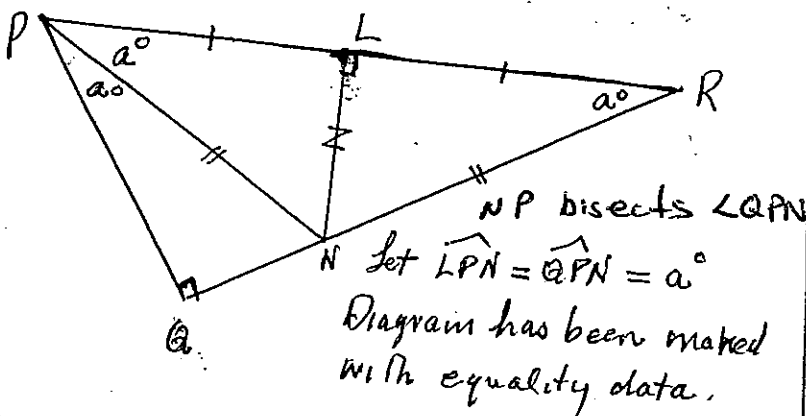
NL is a common side

$\widehat{NLP} = \widehat{NLR} = 90^\circ$ ($NL \perp PR$ given)

$LP = LR$ (L is the midpoint of PR , given)

$\therefore \triangle PNL \equiv \triangle RNL$ (SAS)

(ii)



Well done

MANY DIFFERENT OPTIONS USED. QUITE WELL DONE.

$a = 40$ WITHOUT SOME CORRECT SETTING OUT WAS PENALISED

1 MK AWARDED FOR FINDING ALL THREE PARTS OF PROOF

FOR CORRECT CONGRUENCY RULE

MANY HAD NO REASON OFFERED FOR $\angle QPN = \angle NPL$.

(11) continued.

(a) In $\triangle PQR$ $2a^\circ + a^\circ + 90^\circ = 180^\circ$

$$3a = 90$$

$$a = 30$$

$$\therefore \widehat{NRL} = 30^\circ$$

(b) $\frac{LN}{PL} = \tan 30^\circ = \frac{1}{\sqrt{3}}$

Now since $PL = \frac{1}{2} PR$,

we have: $\frac{LN}{\frac{1}{2} PR} = \frac{1}{\sqrt{3}}$

$$\therefore \frac{LN}{PR} = \frac{1}{2\sqrt{3}}$$

(d) Formula: $(2n-4)$ right-angles

$$\begin{aligned} \text{Angle sum} &= (14-4) \times 90^\circ \\ &= 900^\circ \end{aligned}$$

AGAIN, A NUMBER OF
CORRECT OPTIONS USED.
MOST SUCCESSFULLY.

MANY HAD GOOD SOLN'S
TO THIS.

SOME MADE A SOLID
START BUT COULD NOT
COMPLETE. PART MKS
AWARDED

Well done.

Question 4

(a) (i) $y = \log_e(3 + 2x^2)$

$$\frac{dy}{dx} = \frac{4x}{3 + 2x^2}$$

(ii) $y = (3e^x - 5)^7$

$$\begin{aligned}\frac{dy}{dx} &= 7(3e^x - 5)^6 \times 3e^x \\ &= 21e^x(3e^x - 5)^6\end{aligned}$$

(iii) $y = \frac{\sin 3x}{x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{x \times 3 \cos 3x - (\sin 3x) \times 1}{x^2} \\ &= \frac{3x \cos 3x - \sin 3x}{x^2}\end{aligned}$$

(iv) $y = x\sqrt{x} = x \times x^{\frac{1}{2}} = x^{\frac{3}{2}}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{3}{2} x^{\frac{1}{2}} \\ &= \frac{3\sqrt{x}}{2}\end{aligned}$$

(b) $y = -2x^{-1}$, point $(-1, 2)$

$$\frac{dy}{dx} = 2x^{-2} = \frac{2}{x^2}$$

at $x = -1$, $m = \frac{2}{(-1)^2} = 2$

tangent: $y - y_1 = m(x - x_1)$

$$y - 2 = 2(x + 1)$$

$$y - 2 = 2x + 2$$

$$2x - y + 4 = 0$$

Marks

many forgot formula.

many forgot to differentiate $3e^x$.for finding u' and v'

correct arrangement in formula.

changing to index form ie $x^{\frac{1}{2}}$ differentiating correctly
many used product rule
but this was the long way round.

differentiating correctly

gradient of tangent

for equation of line

alternative: $y = 2x + 4$

$$(c) \lim_{x \rightarrow \infty} \left(\frac{x^2 - 3x + 1}{2x^2 + 5} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{1 - \frac{3}{x} + \frac{1}{x^2}}{2 + \frac{5}{x^2}} \right)$$

$$= \frac{1 - 0 + 0}{2 + 0}$$

$$= \frac{1}{2}$$

1 dividing through by the highest power of x
ie x^2

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2} - \frac{3x}{x^2} + \frac{1}{x^2}}{\frac{2x^2}{x^2} + \frac{5}{x^2}}$$

1 correct answer.

Question 5

(a) $2x^2 + 4x - 1 = 0$

$a = 2, b = 4, c = -1$

(i) $\alpha + \beta = -\frac{b}{a} = -\frac{4}{2} = -2$

$\alpha\beta = \frac{c}{a} = -\frac{1}{2}$

} Both right ✓

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (-2)^2 - 2(-\frac{1}{2})$

$= 4 + 1$

$= 5$

(b) $x^2 - (k-2)x + (k+1) = 0$

(i) $\Delta = b^2 - 4ac$

$\Delta = [-(k-2)]^2 - 4(1)(k+1)$

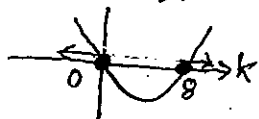
$\Delta = k^2 - 4k + 4 - 4k - 4$

$\Delta = k^2 - 8k$

(ii) $\Delta \geq 0$ for real roots

$k^2 - 8k \geq 0$

$k(k-8) \geq 0$

Ans $k \leq 0$ or $k \geq 8$

Not many problems in this Q.

$$\begin{aligned}
 (c) \quad x^2 - 12x &= 8y - 52 \\
 x^2 - 12x + (-6)^2 &= 8y - 52 + 36 \\
 (x-6)^2 &= 8y - 16 \\
 (x-6)^2 &= 8(y-2) \\
 (x-6)^2 &= 4(2)(y-2)
 \end{aligned}$$

$$(d) \text{ Vertex : } (6, 2)$$

$$\text{Focal length} = 2.$$

$$(\beta) \text{ Focus : } (6, 4)$$

$$(\gamma) \text{ Directrix : } y = 0$$

* If brackets () were not used for coordinates 1 mark was deducted in (d) and (β)

$$(d) (5^x)^2 + 5^x - 2 = 0$$

$$\text{Let } t = 5^x$$

$$t^2 + t - 2 = 0$$

$$(t+2)(t-1) = 0$$

$$t = -2 \text{ or } t = 1$$

$$\therefore 5^x = -2 \text{ N/A as } 5^x > 0 \text{ for all } x$$

$$5^x = 1$$

$$5^x = 5^0$$

$$\underline{x = 0}$$

Question 6

(a) (i) $\int \frac{1}{4x^4} dx$
 $= \int \frac{1}{4} x^{-4} dx$
 $= \frac{1}{4} \frac{x^{-3}}{-3} + c$
 $= -\frac{1}{12x^3} + c$

✓

(ii) $\int \frac{2x}{x^2+10} dx$
 $= \ln(x^2+10) + c$

✓

(b) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2 x dx$
 $= \left[\tan x \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$
 $= \tan \frac{\pi}{4} - \tan \frac{\pi}{6}$
 $= 1 - \frac{1}{\sqrt{3}}$
 $= \frac{\sqrt{3}-1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}(\sqrt{3}-1)}{3} \text{ or } \frac{3-\sqrt{3}}{3}$

✓

✓

✓

Many incorrectly dealt with this question as a logarithm.

Two issues to deal with...

- (1) $\frac{1}{4}$
- (2) x^{-4}

answered well.

$$\int_a^b \sec^2 x = [\tan x]_a^b$$

exact value leave in this form.

completing the table of values

x	3	5	7
$\frac{x}{\ln x}$	$\frac{3}{\ln 3}$	$\frac{5}{\ln 5}$	$\frac{7}{\ln 7}$
	y_1	y_2	y_3

$$\Gamma \equiv \frac{b-a}{n} [(y_1 + y_3) + 4y_2]$$

$$\Gamma \equiv \left(\frac{7-3}{6} \right) [(2.7307 + 3.5973) + 4(3.1067)]$$

$$\Gamma \equiv 12.5 \text{ c.i.d.p}$$

✓

✓ correctly substituting into formula

NOTE: $\frac{b}{3} = \frac{2}{3}$

$$(d) (i) \quad y = x - 3 \quad \text{--- (1)}$$

$$y = x^2 - 3x \quad \text{--- (2)}$$

$$(2) - (1): x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 1 \text{ or } 3.$$

$$\text{when } x = 1 \quad y = -2$$

$\therefore$ point A is (1, -2)

✓

✓

$$(ii) \quad A = \int_1^3 (x^2 - 3x) dx - \int_1^3 (x - 3) dx$$

$$A = \int_1^3 (x^2 - 4x + 3) dx$$

$$A = \left[\frac{x^3}{3} - 2x^2 + 3x \right]_1^3$$

$$A = \left[(9 - 18 + 9) - \left(\frac{1}{3} - 2 + 3 \right) \right]$$

$$A = \frac{4}{3} \text{ units}^2$$

✓

✓

✓

Area between two curves
is best answered using
the formula: $\int_a^b f(x) - g(x) dx$.

Note: 1 mark deducted if you
had the area as $-\frac{4}{3} \text{ units}^2$.

Question 7

(a) $y = x^3 - 3x^2 + 3x - 1$

$\frac{dy}{dx} = 3x^2 - 6x + 3$

$\frac{d^2y}{dx^2} = 6x - 6$

(i) Let $\frac{dy}{dx} = 0$

$3x^2 - 6x + 3 = 0$

$3(x^2 - 2x + 1) = 0$

$(x-1)^2 = 0$

$x = 1$

hence only one stationary point.

When $x = 1$ $y = 1 - 3 + 3 - 1 = 0$

$\frac{d^2y}{dx^2} = 6(1) - 6 = 0$

This will be a horizontal point of inflection at (1, 0)

Concavity Test:

x	0	1	2
$\frac{d^2y}{dx^2}$	-6	0	6

i.e. $\frac{d^2y}{dx^2} \neq 0$

This confirms inflection.

✓ finding the x -value of the S.P

✓ y -value of S.P

✓ determining nature of S.P

← marks weren't awarded for the concavity test but this should be shown.

(ii) Concave up: $f''(x) > 0$

$6x - 6 > 0$

$x > 1$

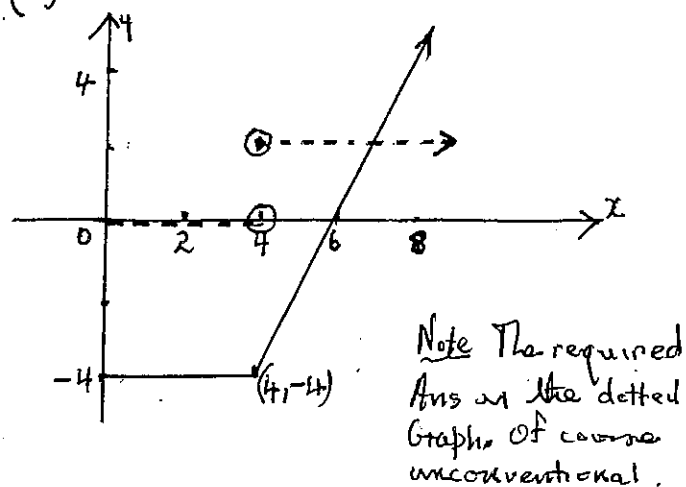
(iii) $f'(x) > 0$
 $(x-1)^2 > 0$

all real x , except $x \neq 1$.

✓

✓

(b)



✓ } lines $y=2$ ~~or~~ and $y=1$

✓ discontinuity at $x=4$

(c) $f''(x) = \cos x + \sin x$

$$f'(x) = \sin x - \cos x + C$$

Now when $f'(0) = 0$, $x = \frac{\pi}{4} \therefore C = 0$

$$\therefore f'(x) = \sin x - \cos x$$

For stationary points $f'(x) = 0$

$$\sin x - \cos x = 0$$

$$\sin x = \cos x$$

$$\tan x = 1 > 0$$

In the stated domain

x in 1st and 3rd quadrant

Basic acute angle $= \frac{\pi}{4}$

Hence the other stationary point is at $x = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$

✓

✓

✓

(d) $y = x\sqrt{x+1} = x(x+1)^{\frac{1}{2}}$

$$\frac{dy}{dx} = 1 \times (x+1)^{\frac{1}{2}} + \frac{1}{2}(x+1)^{-\frac{1}{2}} \times 1 \times x$$

$$= \sqrt{x+1} + \frac{x}{2\sqrt{x+1}}$$

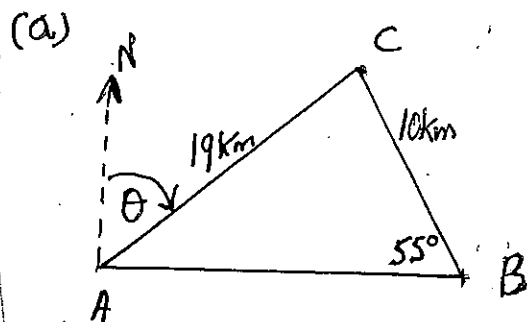
$$= \frac{2(x+1) + x}{2\sqrt{x+1}}$$

$$= \frac{3x+2}{2\sqrt{x+1}}$$

✓ correct application of product rule.

✓ simplification.

Question 8



using The sine rule :

$$\frac{\sin \hat{BAC}}{10} = \frac{\sin 55^\circ}{19}$$

$$\sin \hat{BAC} = \frac{10 \sin 55^\circ}{19}$$

$$\hat{BAC} = 25^\circ 32'$$

True bearing of C from A :

$$\theta = 90^\circ - 25^\circ 32' = 64^\circ 28'$$

ie T 064°28'

(b) $v = \frac{dx}{dt} = 2 - 4 \sin t$

(1) The particle is at rest when $v = 0$

$$2 - 4 \sin t = 0$$

$$\sin t = \frac{1}{2}$$

Now considering the domain $0 \leq t \leq 2\pi$

$\sin t > 0 \therefore t$ in 1st, 2nd. quads

Basic acute angle = $\frac{\pi}{6}$

$\therefore$ Particle at rest when $t = \frac{\pi}{6}$ sec

and when $t = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ sec

True bearing is written in 3 digits.

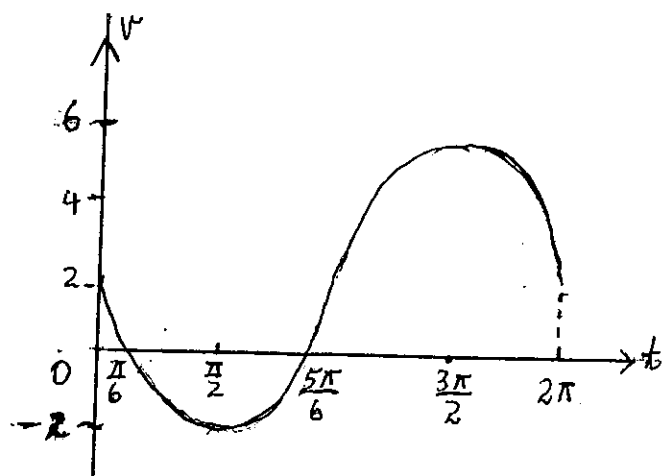
Many missed out on that (Did not penalise this time)

Bonus $\frac{\pi}{6}$, $\frac{5\pi}{6}$ had to be there for 1 mark.

1 right No mark

more than 2 answers no mark

(ii)



Points of intersection
have to be shown on the
t-axis ie $\frac{\pi}{6}$, $\frac{5\pi}{6}$

✓

- (iii) Maximum velocity occurs
when $\sin t$ is a minimum
obviously $-1 \leq \sin t \leq 1$
This occurs at $t = \frac{3\pi}{2}$ ie $\sin t = -1$
 $\therefore$ Max Velocity $= 2 - 4(-1)$
 $= 6 \text{ m/sec}$

Answers from graph
accepted for 2 marks.

- (iv) Now between $t=0$ and $t=\frac{\pi}{2}$.
The particle changes direction
at $t = \frac{\pi}{6}$ (at $v=0$)

$$\begin{aligned} \therefore \text{Total distance travelled} &= \int_0^{\frac{\pi}{6}} (2-4\sin t) dt + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (2-4\sin t) dt \\ &= \left[2t + 4\cos t \right]_0^{\frac{\pi}{6}} + \left[2t + 4\cos t \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left| \left(\frac{\pi}{3} + 4\cos \frac{\pi}{6} \right) - (0 + 4\cos 0) \right| + \left| \left(\pi + 4\cos \frac{\pi}{2} \right) - \left(\frac{\pi}{3} + 4\cos \frac{\pi}{6} \right) \right| \\ &= \left| \frac{\pi}{3} + 4 \times \frac{\sqrt{3}}{2} - 4 \right| + \left| \pi + 4(0) - \frac{\pi}{3} - 4 \times \frac{\sqrt{3}}{2} \right| \\ &= \left| \frac{\pi}{3} + 2\sqrt{3} - 4 \right| + \left| \pi - \frac{\pi}{3} - 2\sqrt{3} \right| \\ &= \frac{\pi}{3} + 2\sqrt{3} - 4 + \left| \frac{2\pi}{3} - 2\sqrt{3} \right| \\ &= \frac{\pi}{3} + 2\sqrt{3} - 4 + \frac{2\pi}{3} + 2\sqrt{3} = \left(\frac{\pi}{3} + 4\sqrt{3} - 4 \right) \text{m} \end{aligned}$$

very few reached the
appropriate end.
Anything close to 3rd
step was awarded
a mark.

Question 9

(a) $P = 500 \times e^{\frac{t}{4}}$

(i) Let $t = 10$

$$P = 500 \times e^{\frac{10}{4}}$$

$$P = 500 \times e^{\frac{5}{2}}$$

$$P \approx 6091$$

∴ After 10 weeks colony has 6091 ants

(ii) $t = ?$ when $P = 10000$

$$500 \times e^{\frac{t}{4}} = 10000$$

$$e^{\frac{t}{4}} = 20$$

$$\frac{t}{4} = \ln 20$$

$$t = 4 \ln 20$$

$$t \approx 11.98$$

∴ Colony would take 11.98 weeks to reach 10,000.

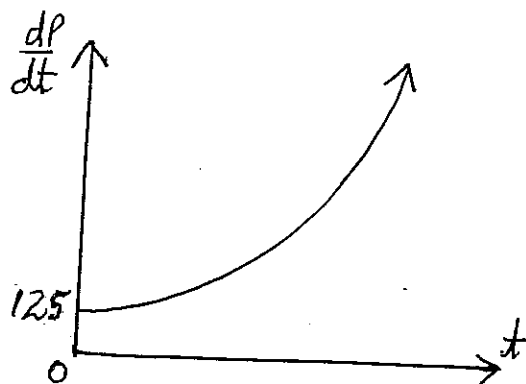
(iii) $\frac{dP}{dt} = \frac{1}{4} \times 500 \times e^{\frac{t}{4}}$

$$= 125 \times e^{\frac{t}{4}}$$

As $\frac{dP}{dt} > 0$ for all t , This means

That the population will always be increasing.

(iv)



generally well done.

1 correct answer

generally well done.

1 correct answer

1 finding $\frac{dP}{dt}$

1 explaining that population is increasing

2 correct shape, vertical intercept + domain.

(b) Loan Amount = \$20000
Monthly Instalment \$399 over 5 yrs
Interest 7.2% p.a compounded monthly.

$$(i) \text{ Interest} = (\$399) \times (12 \times 5) - \$20000 \\ = \$3940$$

$$(ii) \text{ Using } A = PR^n \text{ where } R = 1 + \frac{r}{100} \\ A_1 = 20000(1.006)^1 - 399 = \$19721$$

$$(iii) A_2 = A_1(1.006)^1 - 399$$

$$A_2 = [20000(1.006) - 399]1.006 - 399$$

$$A_2 = 20000(1.006)^2 - 399(1 + 1.006)$$

(iv) continuing from part (iii) and following the pattern, apparent:
Then after 'n' payments the amount A_n owing is:

$$A_n = 20000(1.006)^n - 399(1 + 1.006 + \dots + 1.006^{n-1})$$

↑
This is a GP with n terms, $a = 1$, $r = 1.006$
and using $SA = \frac{a(r^n - 1)}{r - 1}$

$$\therefore A_n = 20000(1.006)^n - 399 \left[\frac{1(1.006^n - 1)}{1.006 - 1} \right]$$

$$A_n = 20000(1.006)^n - 66500(1.006^n - 1)$$

$$A_n = 20000(1.006)^n - 66500(1.006^n) + 66500$$

$$A_n = 66500 - 46500(1.006)^n$$

(v) The One payment = A_{24} .

$$A_{24} = 66500 - 46500(1.006)^{24}$$

$$A_{24} = \$12820.99$$

$$\text{or } \$12821.$$

Simplest part of Q9 worst done.

had to show
 $20000 \times 1.006 - 399$

had to show this step or equivalent.

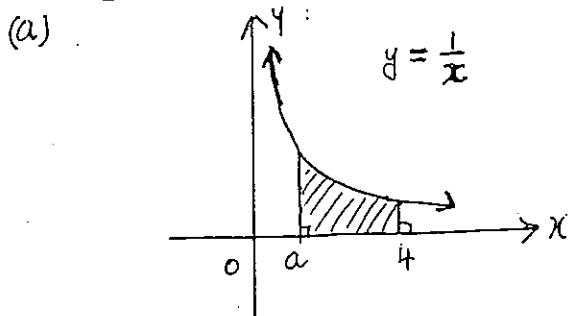
showing the expression for A_n

many people faked from here to answer.

must clearly show arithmetic.

1. many forgot that
24 months = 2 yrs.
Simple substitution into formula was given.

Question 10



$$V = \pi \int_a^b y^2 dx$$

$$\frac{\pi}{2} = \pi \int_a^4 \frac{1}{x^2} dx$$

$$\frac{1}{2} = \int_a^4 x^{-2} dx$$

$$\frac{1}{2} = \left[\frac{x^{-1}}{-1} \right]_a^4$$

$$\frac{1}{2} = - \left[\frac{1}{x} \right]_a^4$$

$$\frac{1}{2} = - \left[\frac{1}{4} - \frac{1}{a} \right]$$

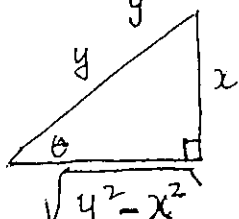
$$\frac{1}{2} = \frac{1}{a} - \frac{1}{4}$$

$$\frac{1}{a} = \frac{3}{4}$$

$$a = \frac{4}{3} = 1\frac{1}{3}$$

(b)

$$\begin{aligned} x \sec \theta &= y \tan \theta \\ \frac{x}{\cos \theta} &= \frac{y \sin \theta}{\cos \theta} \\ x &= y \sin \theta \\ \sin \theta &= \frac{x}{y} \end{aligned}$$



⊥ For correct integral.

Too many tried to use logs.

⊥ For correct substitution

⊥ For correct answer if all else correct

⊥ For correct discovery of $x = y \sin \theta$

⊥ For discovery at $\triangle$.

Now Prove $\tan \theta \sec \theta = \frac{xy}{y^2 - x^2}$
using data from The triangle

$$\text{LHS} = \tan \theta \sec \theta$$

$$= \frac{x}{\sqrt{y^2 - x^2}} \times \frac{y}{\sqrt{y^2 - x^2}}$$

$$= \frac{xy}{\sqrt{y^2 - x^2}}$$

$$= \text{RHS.}$$

(c) $y = mx$, $y = e^{\frac{x}{2}}$

Let (x_0, y_0) be the point of tangency.

$$y = e^{\frac{x}{2}}$$

$$y' = \frac{1}{2} e^{\frac{x}{2}}$$

$$\therefore m = \frac{1}{2} e^{\frac{x_0}{2}} \quad \text{--- (1)}$$

Solving $y = mx$ and $y = e^{\frac{x}{2}}$

$$mx = e^{\frac{x}{2}}$$

$$\text{So } mx_0 = e^{\frac{x_0}{2}}$$

$$\text{Hence } \frac{1}{2} e^{\frac{x_0}{2}} \times x_0 = e^{\frac{x_0}{2}} \quad \text{from (1)}$$

$$\frac{1}{2} x_0 = 1$$

$$x_0 = 2$$

$$\therefore m = \frac{1}{2} e^{2 \times \frac{1}{2}}$$

$$m = \frac{e}{2}$$

Many methods tried.
Credit given for CORRECT
Substitution.

Too many assumed
 $\tan \theta \sec \theta = \frac{xy}{y^2 - x^2}$.

1mk for correct subs
& evaluation.

1 For correct derivative

1 For attempt to solve
two equations simultaneously

1 Correct evaluation at
M.

(d) $S_n = n^2 + 6n$

$$T_n = S_n - S_{n-1}$$

$$= n^2 + 6n - [(n-1)^2 + 6(n-1)]$$

$$= n^2 + 6n - [n^2 - 2n + 1 + 6n - 6]$$

$$= n^2 + 6n - n^2 + 2n - 1 - 6n + 6$$

$$= 2n + 5$$

$$T_n - T_{n-1} = 2n + 5 - [2(n-1) + 5]$$

$$= 2n + 5 - [2n - 2 + 5]$$

$$T_n - T_{n-1} = 2n + 5 - 2n - 2 + 5$$

$$T_n - T_{n-1} = 2 \quad \text{thus 'any term'}$$

minus the term before it is constant

Hence an AP.

1 $T_n = S_n - S_{n-1}$

Most assumed an AP
existed.

1 $T_n = 2n + 5$

Many tried to find T_1, T_2, T_3
etc

1 $T_n - T_{n-1} = d$

$$d = 2.$$